

THE NEXT PROBLEMS TO SOLVE IN AUGMENTED REALITY

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Abstract—Augmented reality (AR) is a growing phenomenon. We are on the verge of ubiquitously adopting Augmented Reality (AR) technologies to enhance our perception and help us see, hear, and feel our environments in new and enriched ways. AR will really change the way people see the universe. This paper provides a review on classification of different augmented reality challenges ranges from human factors to hard problem of law and policy which aims to explore new challenges and issues for the acceptance of AR technology.

Keywords—Augmented Reality, Augmented Reality technologies, challenges, AR acceptance

I. INTRODUCTION

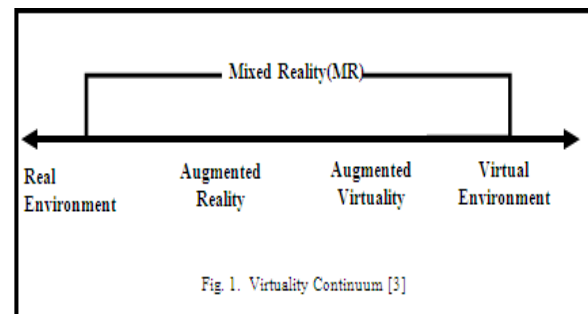
A vision for augmented reality (AR) has been around for decades. Augmented reality is a technology in which computer generated objects or elements are used to augment the view of real world environment. Augmented reality has been around since the 1960s. As AR takes ‘reality’ as a starting point, then adding to it so we can say that it is a term more strongly related to the term ‘reality’ than to the term ‘Virtual reality’. AR is related to ‘mediated reality’ between Virtual Environments and Real world (Telepresence) [1]. Now, with the emergence and dispersal of the Mobile AR and AR browsers, people start to accept this different and exciting character of human computer interaction. AR has gained more research interest and a focus recently.

Augmented reality (AR) and virtual reality (VR) tend to be fields that have a mass of differences; the user is entirely immersed in artificial world in virtual reality (VR) whereas in augmented reality (AR) a computer is used to add cues to the already existing real world. In augmented reality user can interact with real world and can never cut off from the real

world. In contrast the users of virtual reality completely immersed in artificial world [1]. AR not only adds items in real world but also represents useful digital information in real world [2].

II. DEFINING TECHNOLOGY VIRTUAL, MIXED AND AUGMENTED

Augmented Reality (AR) can be define as a real-time direct or indirect view of a physical real world environment that has been augmented by adding virtual computer generated information to it like video, sound, graphics or GPS. In augmented reality, the user interacts with the real world in a natural way. By contrast, virtual reality replaces the real world with a simulated one. The augmented reality (AR) is closer to the real world environment and augmented virtuality (AV) is nearer to a virtual world. Milgram’s Reality-Virtuality Continuum is defined as it covers the real world and the virtual world that are composed of both AR and AV [3].



More formally, augmented reality is also defined by involving three characteristics [1]

- the actual combination-state involving virtual and real-world
- interactive in real environment
- Authorized virtual and real objects in 3D

These three characteristics provide a proper definition of augmented reality.

AR adds computer generated items in real world as well as also represents useful digital information in real world. Augmented reality boosts the sense of reality through laying virtual items over the real world in real time [4]. AR is related to a mediated reality, in which a computer system is used to modify view of reality.

Recently AR has gained more research focus; several meanings of AR are concluded by researchers. AR is starting a new mode of interaction with the environment and enhances a wide spectrum of human activities economic, cultural, and social etc. Augmented reality is now quite well consolidated and supports the users in a wide range of applications. AR technology, such as the Google Glass project [5] and contact lenses by Innovega are opening exciting scenarios [6].

The main components of augmented reality are displays, tracking technologies, interfaces, registration systems, hardware and software. Augmented reality technologies, which are rapidly evolving and becoming commercially available, will create new challenges and limitations. Despite the growing interest in AR and the large body of advances and research, several challenges and issue still exist and need to be addressed.

III. CHALLENGES AND LIMITATIONS OF AUGMENTED REALITY

Augmented reality (AR) is a well-known technology that can be utilized to provide mass-market users a valuable and customizable support in a large spectrum of personal applications, by overlapping computer-generated hints to the real world. One factor that influences the success of a new technology is the acceptance by users. As this field is expected to increase in importance rapidly due to technological advancements and research into functionality, acceptance and usefulness, it is important to identify what the basic limitations for AR acceptance by users. The aim of this paper is therefore mainly concerned with the challenges, limitations and issues that rise in the acceptance of augmented reality by users.

These can be broadly divided into different categories which are discussed as below:

A. Social Acceptance Challenges

Getting people to use AR may be more challenging than expected, and many factors play a role in social acceptance of AR ranging from unobtrusive fashionable appearance (gloves, helmets, etc.) to privacy concerns. These

fundamental issues must be addressed before AR is widely accepted.

a) Natural Interaction

For social acceptance the user should have the ability to interact with the system in a usual natural mode. The unnatural interaction between the system and user may create embarrassing situation in public places, E.g If a person's phone receives a message when he/she is in a public place it not only disrupts the owner but also the other persons [2]. Augmented Reality systems develop for mobile phones need to be subtle, unobtrusive. Indeed, the main problem with social acceptance comes from the level of disruption portable devices create in public places and during conversations.

b) Fashion Acceptance

To fill the gap from laboratories to the commercial use of mobile AR system, there is a need to overcome the fashion issues related to wearing HMD or other transparent devices that users not want to wear. Therefore, the mobile developers need to notice the fashion trends as it may appear a big hurdle to jump [4].

B. Technical Challenges

a) Hardware problems

The hardware used should be small, light, and easily portable and fast enough to display graphics. Also the battery life used by these complicated AR devices is another limitation for AR's uses. Furthermore, there is an issue of camera quality in mobile device. The imaging produce is sometimes poor due to bad lighting and color fidelity [7]. Another issue is Sensor accuracy like GPS, compass etc.

C. Visualization challenges

Visualization challenges include display issues like contrast, resolution, brightness, and field of view. Another visualization issue is occlusion, i.e. a process which determines which surface or its parts are not visible from a certain viewpoint [8]. For a realistic view, correct handling of occlusion between virtual objects and real world objects in the scene is important [9].

a) Display Devices problems

TABLE 1. AR DISPLAY PROBLEMS

Problems in Augmented Reality Displays	
Insufficient brightness	Wide Field of View
Camera's Improper resolution	Projection issues
Different field of view	Fatigue and eye strain
Distortions	color depth
Delay/latency	luminance
Line of sight	focus depth

Display	Visualization issues
Real objects are not completely occlude by virtual objects	system alignment is inaccurate for eye
Low brightness in Optical and video see-through displays	Unsuccessful user interface interaction
system alignment is inaccurate for eye	Unsuccessful user interface interaction

D. Tracking challenges

Tracking is another important issue in outdoor AR. Tracking in unprepared environments remains a challenge. Some factors involved in most augmented reality technologies are latency [10], noise and orientation errors on tracking distant objects in augmented reality [11]. Most portable computers have only one CPU which limits the amount of visual and hybrid tracking.

E. Recognition Challenges

A different problem for augmented reality objects is the recognition of the boundaries between physical world and the virtual world [12].

F. Environmental issues

Another problem is that a lot of environments need to be prepared for working with augmented reality. Markings are needed for the augmented reality system to pick up certain locations [13]. Some other environmental factors includes

- Lighting and weather conditions
- The color scheme and diversity of an environment
- Camera quality and handling in bad lighting conditions
- Color fidelity in outside environments

G. UX challenges of AR

This type of augmented reality limitations are totally depend on user or either individual users

a) User Interface Limits

Different types of interfaces are used in augmented reality depending upon the type of AR application which causes different problems like

- Data availability in an unsuccessful way.
- User's reports and queries not organized in valuable way.

b) Interaction challenges

Interaction challenges refer to the interaction of users with virtual and real objects at the same time. Interaction uses various interfaces that may be acoustic, haptic, tangible, gaze, or text-based through

which the user interacts with virtual objects. Interaction Techniques and User Interfaces are still problems which need to be solved [14].

c) Overload and over-reliance in Interfaces

User interface must also follow some guidelines as not to overload the user with information while also preventing the user to overly rely on the AR system such that important cues from the environment are missed [15].

d) Human Factors

These challenges are completely depend on individual users like

- Weak Usability
- Adaptation
- Fatigue and eye strain

H. Privacy Challenges

Privacy is one of the serious challenges in AR acceptance that really needs to overcome

- Location of user
- Confidential history of user especially in medical AR systems.

There are also privacy issues. With improvement in image-recognition software combined with AR technology it allows user to point their phones at other people, even strangers, and straightaway we can see information from their Facebook, Twitter, LinkedIn or other online profiles. While most of these apps people willingly to put their information online, it may be an unwelcome and unpleasant to meet someone, who knows so much about your life and background instantly.

I. Registration challenges

a) Latency or system delays

A large source of dynamic registration errors are system delays [16]. Techniques like precalculation, temporal stream matching (in video see-through such as live broad-casts), and prediction of future viewpoints may solve some delay.

b) Depth perception

One difficult registration problem is accurate depth perception. Some additional problems including accommodation vergence conflicts or low resolution and dim displays cause object to appear further away than they should be [17].

c) Alignment challenges

Alignment is concerned with a proper placement of a virtual object to the real world objects. Incorrect alignment may cause problems such as incorrect rendering of information to the real world. This

misalignment is more severe in medical applications. Alignment challenges include registration problems, which is the most basic problem in AR [1].

J. Performance challenges

Performance issue is a major concern of mobile AR. These challenges are concerned with real time processing, responding and evolving with the change of real world environment. Real time processing can slow down the performance of augmented reality applications [18].

K. Mobility Challenges

These challenges are concerned with the portability of augmented reality systems. It should be small and light so it can be used any -where. The best augmented reality system will be portable outside a controlled environment [1].

L. Portability and outdoor use

Optical and video see-through displays are usually unsuited for outdoor use due to low brightness, contrast, resolution, and field of view.

M. Usefulness

Most mobile AR application nowadays, only focus on the technology itself and it lacks the user experience elements, which has make it useless.

CONCLUSION

While augmented reality technology is still new, it has already proved to be useful in many fields. As a new interaction medium, AR requires strong understanding of the users, expectations and needs. It is important to get it right and make sure it adds value to the user experience.

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